

Pentagonal Number Theorem

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1 Partitions

Definition 1. A *partition* of a positive integer n , also called an integer partition, is a way of writing n as a sum of positive integers. Two sums that differ only in the order of their summands are considered the same partition.

The *partition function* $p(n)$ represents the number of possible partitions of a natural number n .

Example 2. $5 = 4 + 1 = 3 + 2 = 3 + 1 + 1 = 2 + 2 + 1 = 2 + 1 + 1 + 1 = 1 + 1 + 1 + 1 + 1$
 $p(5) = 7$

Lemma 3. The generating function for $p(n)$ is given by

$$\sum_{n=0}^{\infty} p(n) x^n = \prod_{k=1}^{\infty} (1 + x^k + x^{2k} + \dots) = \prod_{k=1}^{\infty} (1 - x^k)^{-1}.$$

Definition 4. For a positive integer n we denote by $p_e(n)$ the number of partitions of n into an even number of *different* parts, analogously, we denote by $p_o(n)$ the number of partitions of n into an odd number of *different* parts.

Lemma 5. We have

$$\prod_{k=1}^{\infty} (1 - x^k) = \sum_{s=0}^{\infty} \sum_{k_1 < k_2 < \dots < k_s} (-1)^s x^{k_1 + k_2 + \dots + k_s} = \sum_{n=0}^{\infty} c_n x^n$$

where $c_0 = 1$ and for $n \geq 1$ and $c_n = p_e(n) - p_o(n)$.

Example 6. We compute $\prod_{i=1}^{20} (1 - x^i) + o(x^{20})$.

The answer is $1 - x - x^2 + x^5 + x^7 - x^{12} - x^{15} + O(x^{20})$.

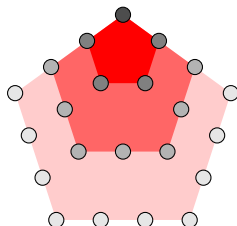
n		$p_o(n)$	$p_e(n)$
1	1	1	0
2	2	1	0
3	$3 = 2 + 1$	1	1
4	$4 = 3 + 1$	1	1
5	$5 = 4 + 1 = 3 + 2$	1	2
6	$6 = 5 + 1 = 4 + 2 = 3 + 2 + 1$	2	2
7	$7 = 6 + 1 = 5 + 2 = 4 + 3 = 4 + 2 + 1$	2	3

2 Pentagonal number theorem

Theorem 7 (Euler).

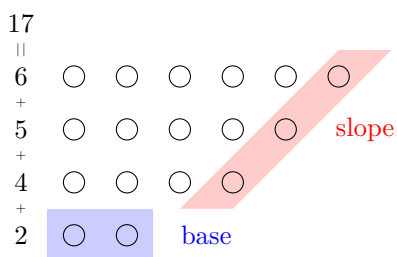
$$\prod_{i=1}^{\infty} (1 - x^i) = 1 + \sum_{k=1}^{\infty} (-1)^k (x^{(3k^2-k)/2} + x^{(3k^2+k)/2}) = \sum_{k \in \mathbb{Z}} (-1)^k x^{(3k^2-k)/2}.$$

Remark 8 (Pentagonal numbers). $\frac{k}{\frac{3k^2-k}{2}} \left| \begin{array}{c} -3 \\ 15 \end{array} \right| \frac{-2}{7} \left| \begin{array}{c} -1 \\ 2 \end{array} \right| \frac{0}{0} \left| \begin{array}{c} 1 \\ 1 \end{array} \right| \frac{2}{5} \left| \begin{array}{c} 3 \\ 12 \end{array} \right|$



2.1 Proof

Consider a diagram of a partition of n into different parts.



We define the following two operations on the diagrams and denote them α and β .

α : If $|b| \leq |s|$ and if b and s have no common point, or if $|b| \leq |s| - 1$, then we move b up and turn its points into new slope.

β : If $|b| > |s|$ and if b and s have no common point, or if $|b| \geq |s| + 2$, then we move the slope down and turn its points into the new base.

Remark 9. If b and s have a common point and $|b| = |s|$ or $|b| = |s| + 1$ then neither α nor β can be applied.

α



β



To each diagram D we can apply at most one of the operations α or β . If α can be applied to D then β can be applied to $\alpha(D)$. If β can be applied to D then α can be applied to $\beta(D)$. Thus we have a 1:1 correspondence

$$\left\{ \begin{array}{l} \text{odd partitions} \\ \text{into distinct parts} \\ \text{s. t. } \alpha \text{ or } \beta \text{ can be applied} \end{array} \right\} \leftrightarrow \left\{ \begin{array}{l} \text{even partitions} \\ \text{into distinct parts} \\ \text{s. t. } \alpha \text{ or } \beta \text{ can be applied} \end{array} \right\}$$

When can neither α nor β be applied?

- b and s have a common point and $|b| = |s| = k$

$$n = k + (k + 1) + \dots + (2k - 1) = \frac{3k^2 - k}{2}.$$

- b and s have a common point and $|b| - 1 = |s| = k$

$$n = (k + 1) + \dots + 2k = \frac{3k^2 + k}{2}.$$

Therefore:

$$p_e(n) - p_o(n) = c_n = \begin{cases} (-1)^k, & \text{if } n = \frac{3k^2 \pm k}{2} \\ 0, & \text{otherwise.} \end{cases}$$

This finishes the proof.

3 Jacobi triple product

Theorem 10. For all $w \neq 0$ and q with $|q| < 1$:

$$\prod_{k=1}^{\infty} (1 - q^{2k}) (1 - q^{2k-1} w^2) (1 - q^{2k-1} w^{-2}) = \sum_{k=-\infty}^{\infty} (-1)^k w^{2k} q^{k^2}.$$

3.1 Proof

Set $\phi_n(w, q) := \prod_{k=1}^n (1 - q^{2k-1} w^2) (1 - q^{2k-1} w^{-2})$. We have

$$\phi_n(qw, q) = \phi_n(w, q) \frac{(1 - q^{2n+1} w^2) (1 - q^{-1} w^{-2})}{(1 - qw^2) (1 - q^{2n-1} w^{-2})} = \phi_n(w, q) \frac{1 - q^{2n+1} w^2}{-qw^2 + q^{2n}}. \quad (1)$$

We consider the following expansion

$$\phi_n(w, q) = \sum_{k=-n}^n A_{n,k}(q) w^{2k}, \quad (2)$$

where $A_{n,k}(q)$ is a polynomial in q . We have a symmetry $A_{n,k}(q) = A_{n,-k}(q)$. We find

$$A_{n,n}(q) = q^{1+3+5+\dots+(2n-1)} = q^{n^2}. \quad (3)$$

We substitute (2) into (1) and obtain

$$A_{n,k}(q) = A_{n,k-1}(q) q^{2k-1} \frac{1 - q^{2n-2k+2}}{1 - q^{2n+2k}}. \quad (4)$$

From (3) and (4) we find

$$A_{n,k} = \frac{q^{k^2}}{(1 - q^2) \dots (1 - q^{2n})} \prod_{s=n-k+1}^n (1 - q^{2s}) \prod_{s=n+k+1}^{2n} (1 - q^{2s}). \quad (5)$$

From (5) we obtain

$$\prod_{k=1}^n (1 - q^{2k}) (1 - q^{2k-1} w^2) (1 - q^{2k-1} w^{-2}) = \sum_{k=-n}^n (-1)^k w^{2k} q^{k^2} B_{n,k}(q).$$

where

$$B_{n,k}(q) = \prod_{s=n-k+1}^n (1 - q^{2s}) \prod_{s=n+k+1}^{2n} (1 - q^{2s}).$$

We have $B_{n,k} = 1 + O(q^{n-k+1})$. We take the limit $n \rightarrow \infty$ and finish the proof.